Institute of Infrastructure Technology Research and Management

End Semester
(Start time: 02.30 p.m. and End time: 05:30 p.m.)
B. Tech. (All Branches)
Semester - VI

Date: 24/04/2023

Course Name: Infrastructure Planning and Management

Course Code : HS 225002

Max. Marks: 100

Instructions:

1. Course instructor may penalize the student by deducting the marks, if answer is adjudged to be copied during assessment. This act shall be recorded as UFM (unfair means).
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Section A

Attempt all the questions. Each question carries 10 marks. (10X 2 = 20 Marks)

Q.1) Discuss the relevance of time value of money concept in the evaluating infrastructure projects with example.

Q.2) What are the various risks associated with infrastructure projects, and what strategies can be implemented to mitigate them?

Section B

Attempt all the questions. Each question carries 10 marks. (10 X 3 = 30 Marks)

Q.1) L & T infrastructure is planning to undertake a road infrastructure project with initial investment of ₹10,00,000. Consider the following cash flow from the project and suggest whether L & T Infrastructure should invest at 18% and 15% discount rate. Why and why not?

Year	1	2	3	4	5
Cashflow	2,00,000	3,00,000	4,00,000	3,00,000	2,00,000

Q.2) Adani Infrastructure is using both debt and equity financing to fund their infrastructure project, with debt financing accounting for 80% of the total. Based on the following information, calculate the weighted average cost of capital (WACC)

1. Risk free interest rate: 8%;
2. Equity beta : 1.2 ;
3. Market Premium : 8.4%;
4. Pre-tax cost of debt : 10%;
5. Income tax rate : 35%

Q.3) Reliance Infrastructure Ltd plans to construct a toll road on a BOT model, with an initial investment of ₹ 10,00,000. Expert analysis suggests that the probability of India's economic conditions being good, ok, or bad is 0.25, 0.45, and 0.3, respectively. Depending on the traffic volume, the expected revenues for Reliance Infrastructure are ₹ 14,00,000 for high traffic, ₹ 12,00,000 for medium traffic, and ₹ 5,00,000 for low traffic. Based on this information and table mentioned below, determine whether investing in the project is viable or not using decisioneering approach.

Economy vs Traffic	Good	OK	Bad
High	0.6	0.4	0.1
Medium	0.3	0.45	0.3
Low	0.1	0.15	0.6

Section C

Attempt all the questions. Each question carries 10 marks. (10 X 5 = 50 Marks)

Q.1) What are the challenges associated with managing electrical infrastructure in India? Illustrate your answer with the help of the Dabhol power plant case?

Q2) Evaluate the opportunities and constraints present in the infrastructure management of the Indian aviation industry during AirAsia's entry into the country? Additionally, what were the challenges and issues that AirAsia India encountered during this time?

Q.3) Explain the various legal and environmental challenges that may impact infrastructure projects, using the Lavasa project as an example to support your reasoning?

Q.4) Define smart infrastructure and provide an explanation of the Smart City Mission implemented by the government?

Q.5) Describe the evolution of the Indian telecom industry after new telecom policy and compare the revenue models of Airtel and Jio? Furthermore, what have been the effects of Jio's entry on both customers and competitors?